

WELL COMPLETION REPORT (SYNTHETIC)

OIL-HPJ-004 - Hapjan-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000043
Field / District	Hapjan / Tinsukia
Block	OALP-III-07
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2001-05-20 / 2001-07-11
Measured / TVD	2819 m / 2781 m
Trajectory	Vertical; inc 2.9 deg; az 9.3 deg
Formation Interval	1232-2124 m
Lithology	Interbedded sandstone and shale
Pore Pressure Reference	2770 psi
Fracture Gradient	15.37 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (53.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	704	Surface
9 5/8 in casing	1635	Intermediate
7 in liner/casing	2480	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2001-06-01 804 m	Stuck Pipe	Critical	High drag was recorded while circulating remained possible. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2001-05-27 537 m	Cementing Issue	High	Cement density/returns trend indicated placement uncertainty. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Offset loss history should inform cementing hydraulics.
2001-05-29 660 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2001-05-30 684 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2001-06-05 1015 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2001-06-30 2263 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2001-06-30 2285 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 2819 m MD in the Barail interval. The NWIS knowledge record links 13 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.